Dual-Stream Discriminative Attention Network for Cross-Scene Hyperspectral Image Classification

Chenglong Wang, Yi Guo, and Jiaojiao Fu

***Abstract*— In hyperspectral image (HSI) classification, the challenge of the small-sample-size problem persists as a significant obstacle due to the high cost of labeling samples. To effectively train models with a limited sample set, the application of a transfer learning approach called cross-scene HSI classification is considered a viable solution to address this problem. In cross-scene HSI classification, a source scene with sufficient labeled samples is leveraged to assist in classifying a target scene that lacks labeled samples. Considering that real HSIs may be captured by different sensors, we propose a novel heterogeneous transfer learning algorithm called dual-stream discriminative attention network (DSDAN) to address the task of cross-scene HSI classification. The DSDAN predominantly comprises three pivotal modules: 1) a dual-stream lightweight hybrid CNN (DSLHC) incorporates both the source stream and the target stream and is applied to extract alignment spatial–spectral features from heterogeneous data; 2) a discriminative attention block (DAB) is created to address the domain shift between two scenes. Following the DSLHC, the DAB assigns discriminative attention weights to the source features, facilitating a closer alignment of features from two scenes; and 3) a specially designed cross-domain loss (CDL) is designed to drive intraclass samples from two scenes to become more consistent, while interclass samples from two scenes become more distinct, thereby further mitigating domain shift. By combining DSLHC, DAB, and CDL, the complete DSDAN model is established. The effectiveness of DSDAN is validated using three real cross-scene HSI datasets.**

***Index Terms*— Cross-domain discriminative attention mechanism, cross-domain loss (CDL), cross-scene classification, hyperspectral image (HSI).**

I. Introduction

HYPERSPECTRAL images (HSIs) are multidimensional images captured by specialized sensors. Unlike traditional RGB images, HSIs provide rich spectral dimensionality, often spanning from tens to hundreds of spectral bands. Given the rich information content of HSIs, HSIs are garnering increasing attention [1], [2]. Currently, HSIs have found applications in many fields, such as crop identification [3], oil spill mapping [4], medical tumor diagnosis [5], industrial food packing [6], collaborative classification [7], and so on.

However, one challenge in utilizing HSIs is the need for a large number of labeled training samples, which can be costly and time-consuming to acquire. Considering that different scenes may share the same land covers, and the co-occurrence land covers are interconnected [8], cross-scene HSI classification [9] is proposed as a solution. In detail, one scene with an ample supply of labeled samples (referred to as the source scene) assists another scene with only a limited number of labeled samples (referred to as the target scene) in its classification process.

In cross-scene HSI classification, the dissimilarity between different scenes constitutes a primary challenge for cross-scene classification. Typically, due to variations in sensor acquisition, geographical locations, weather conditions, and so on, the data sizes and distributions of land covers within two scenes are distinct [10], [11]. This phenomenon is commonly termed spectral shift or data shift [12]. The primary objective in addressing cross-scene classification is the minimization of these variances, that is, the aim is to render samples of the same class between different scenes as similar as possible while ensuring that different classes remain as distinct as feasible. In the early studies of cross-scene HSI classification, researchers commonly explored methods to transform samples from the source scene by mapping them to the target scene to mitigate domain shift effects. Shen et al. [13] introduced a domain adaptation technique called hyperspectral feature adaptation and augmentation (HFAA), which leverages two distinct projection matrices to learn a shared subspace, enhancing the similarity of samples with the same labels in both domains. Ye et al. [14] proposed a framework for cross-scene HSI classification that amalgamates the extraction of spatial–spectral features with the learning of a manifold-constrained feature subspace. In this framework, the attainment of a manifold-constrained feature subspace is actualized through the employment of multitask nonnegative matrix factorization (MTNMF) [14]. Li et al. [15] created an iterative weighted heterogeneous transfer learning framework that iteratively discerns the common space between source and target data. This iterative weighting strategy is applied to source samples and culminates in the achievement of cross-scene classification [15]. Nevertheless, it is noteworthy that the majority of these methods typically necessitate a substantial

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quantity of labeled samples for training, which may not be congruent with the practical situation.

To tackle the dependency on a large number of labeled samples, few-shot learning (FSL) has emerged as a preferred method in cross-scene HSI classification. The widespread adoption of FSL stems from its notable capacity to adjust to heterogeneous transfer scenes. By acquiring knowledge of features with discriminative structures, FSL effectively mitigates differences arising from variations in spectral dimensions between different scenes, thereby significantly enhancing the model's domain adaptation capability. Zhang et al. [16] introduced a dual graph cross-domain FSL (DG-CFSL) incorporating a domain alignment strategy. Liu et al. [17] developed a deep FSL model to address the challenges of cross-scene HSI classification with limited labeled samples. Additionally, Zhang et al. [18] proposed a graph information aggregation cross-domain FSL approach (Gia-CFSL). However, despite the significant achievements of FSL in limited sample scenes, it still exhibits some limitations. FSL heavily relies on the initial source training samples, overlooking the presence of a small number of labeled samples in the target scene. FSL struggles to capture complex relationships between classes and falls short in simultaneously addressing both scenes, often necessitating the formulation of effective transfer strategies.

In recent years, the cross-domain attention mechanism has gained increased prominence as a potential solution to these issues. The cross-domain attention mechanism typically involves learning a set of weights or coefficients. Unlike traditional transfer learning methods, the model can reuse knowledge acquired from one domain while also adaptively selecting and weighing different features. Additionally, the cross-domain attention mechanism can simultaneously consider both the source and target scenes, leveraging the knowledge learned from the source scene to aid in the classification of the target scene. These capabilities enable it to effectively address small-sample-size problems and heterogeneous transfer learning challenges. Wang et al. [19] introduced a graph meta-transfer network (GMTN), which combines graph neural networks and meta-learning strategies. The GMTN leverages a bidirectional propagation mechanism to ascertain feature similarities between the source and target domains. Through the assimilation of local and global image information, along with the incorporation of an attention mechanism, it accentuates vital regions while subduing superfluous feature expressions. Zhu et al. [20] created a feature alignment algorithm utilizing cross-domain spatial–spectral attention and deformable convolution for cross-scene HSI classification. The method fused a deformable convolution strategy, multiple feature alignment, and a cross-domain spatial–spectral approach to effectively capture feature information in images and adapt to classification needs across different scenes. Zhu et al. [21] introduced an attention-based multiscale residual adaptation network (AMRAN). The AMRAN model utilized an attention-based multiscale adaptive methodology for cross-scene classification. Through the alignment of edges and conditional distributions, AMRAN discerned a more comprehensive array of edge and conditionally invariant features. The cross-domain attention mechanism concentrated on local information, proving notably efficacious in aligning conditional distributions between two HSI scenes. This capability enables spatial and spectral attention maps to meticulously direct the transfer of pertinent information.

However, many transfer algorithms still confront the following questions: 1) some of these algorithms impose strict requirements, necessitating samples of the same class and equal quantity between the source and target domains, rendering them less practical for real applications and 2) most of them used homogeneous transfer learning rather than heterogeneous transfer learning, a limitation that does not align with the complexity of many real situations. After taking the advantages and disadvantages of the previous transfer learning algorithms into consideration, we propose an innovative approach called dual-stream discriminative attention network (DSDAN), aimed at achieving enhanced domain adaptation. Unlike other approaches, we combine a feature extraction network with a cross-domain discriminative attention mechanism and further optimize it with a cross-domain loss (CDL) function. The main contributions of DSDAN are as follows.

1) A dual-stream lightweight hybrid CNN (DSLHC) is proposed to conduct feature extraction from the source scene and target scene, respectively. Through the DSLHC, the spatial–spectral features of the two heterogeneous scenes are aligned to the same dimension for subsequent operations, thereby mitigating domain shift to some extent. Moreover, the lightweight hybrid architecture adeptly captures features from high dimensions, addressing overfitting problems in scenes with limited training samples, and demonstrating exceptional compatibility across diverse HSI scenes.
2) A discriminative attention block (DAB) is introduced to assess the goodness of source scene features. Specifically, good source scene features should be as close as possible to the features of the same class in the target scene and as far away as possible from the features of different classes in the target scene. Through a carefully designed update rule, discriminative attention weights can be calculated. These discriminative attention weights are then applied to the source scene features through element-wise multiplication, enabling the source features to be closer to the target features. Leveraging cross-domain attention and dynamic attention adjustment, the flexibility and generalization ability of the model are enhanced. Consequently, domain shift is significantly reduced.
3) Following the DAB, a specially designed CDL is introduced. The CDL aims to improve intraclass consistency and interclass distinctiveness, addressing the imbalance issue between source and target scene samples through a sample weight balancing mechanism. This enhancement contributes to the model's adaptability to the target scene, further constraining domain shift.

The subsequent sections of this article are organized as follows. In Section II, an overview of deep learning in HSI classification and cross-domain attention mechanism

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is provided. The details of the proposed DSDAN are shown in Section III. Next, Section IV is dedicated to presenting the results of our experiments and providing an in-depth analysis. Finally, Section V gives the conclusion of this article.

II. Related Work

The DSDAN we proposed utilizes a cross-domain discriminative attention mechanism to address the domain shift problem between two HSI scenes. This section provides a concise introduction to the application of deep learning in HSI classification and cross-domain attention mechanisms.

A. Deep Learning in HSI Classification

In the early studies of HSI classification, traditional classifiers such as support vector machines (SVMs) [22], [23], k-nearest neighbor (KNN) [24], [25], decision trees [26], [27], and so on were predominantly utilized. However, due to the abundant spectral information present in HSIs, these methods often struggled to extract sufficient knowledge.

In recent years, deep-learning approaches have gained increasing prominence in HSI classification due to their outstanding feature extraction capabilities [28], [29], [30], [31]. Zhou et al. [32] used a stacked autoencoder to classify HSIs. By learning compact and discriminative feature representations, this method can effectively classify pixels in HSIs. Zhong et al. [33] conducted a study on the application of deep belief networks (DBN) to HSI classification and introduced a learning approach aimed at enhancing network diversity. This method introduced randomness during training to enable the DBN to learn more diverse and rich feature representations on different training samples. Chen et al. introduced an innovative spectral representation method called ISSFSP [34], which captures both spectral and spatial information through HSI transformation. By utilizing CNN for training, this method can achieve improved classification accuracy and better spatial localization performance. Zhou et al. [35] introduced a multiscanning-based RNN–Transformer model for HSI classification, effectively amalgamating the strengths of both recurrent neural networks (RNNs) and Transformers to make optimal use of spatial and spectral information.

Nevertheless, it is crucial to recognize that a significant portion of deep-learning algorithms typically necessitate a substantial volume of training samples to achieve optimal model efficiency, which may not be feasible in real situations. Consequently, there is a growing trend toward the application of deep-learning methods in the domain of cross-scene HSI classification [36], [37], [38]. For instance, Zhang et al. [39] have introduced an innovative approach that combines unsupervised domain adaptation techniques with classification tasks, incorporating a language-aware module designed to extract semantic information from textual labels, thereby enhancing support for image classification. Similarly, Zhao et al. [40] proposed an unsupervised domain adaptation method focused on learning feature similarity between different scenes. This method involves the integration of multiscale feature fusion and decision boundary acquisition to more effectively utilize multilevel feature information, resulting in improved classification accuracy. Cheng et al. [41] ventured into the domain of causal meta-transfer learning for cross-domain few-shot HSI classification. This method leverages meta-learning strategies to discern causal relationships between domains using a limited number of samples, ultimately achieving high-precision classification results.

These pioneering endeavors underscore the increasing importance of harnessing deep-learning techniques for cross-scene HSI classification, particularly in situations where collecting a substantial number of labeled samples is impractical or cost-prohibitive.

B. Cross-Domain Attention Mechanism

The cross-domain attention mechanism represents an extension of the self-attention mechanism, typically used for handling interdomain information interaction and feature fusion. The primary objective of this mechanism is to enable the model to effectively leverage the knowledge and features learned in one domain within another domain. The cross-domain attention mechanism was initially introduced in the Transformer model for machine translation tasks [42]. Over recent years, an increasing number of researchers have integrated cross-domain attention modules (CDAMs) into diverse research areas, such as recommender systems in online plantforms [43], language recognition [44], remote sensing classification [45], and so on. For example, Zhang et al. [46] introduced a consistency penalty to ensure the effective operation of the CDAM for crowd counting. Zhang et al. [47] created an interactive attention transfer network (IATN), which can seamlessly integrate information from both sentences and aspects, effectively influencing the ultimate sentiment representation. Ma et al. [48] proposed a novel approach called mutually boosted attention transformer (MBATrans), aimed at addressing domain shifts across diverse scenes. MBATrans incorporates an innovative mutual boosted attention (MBA) module, in conjunction with a GAN, to enhance the discriminative capabilities of the model.

Inspired by the aforementioned methods, we construct a cross-domain DAB incorporating DSLHC to tackle the challenge of cross-scene HSI classification. The details of DSDAN will be discussed in Section III.

III. Proposed DSDAN

In cross-scene HSI classification, the source training data $\mathbb{D}_{\mathcal{S}}^{i}, i \in 1, 2, \ldots, n_{\mathcal{S}}$ are sufficient while the target training data $\mathbb{D}_{\mathcal{T}}^{j}, j \in 1, 2, \ldots, n_{\mathcal{T}}$ are rare. Here, $\mathcal{S}$ and $\mathcal{T}$ represent samples from the source and target scenes, respectively. $n_{\mathcal{S}}$ signifies the number of training samples within the source scene, and $n_{\mathcal{T}}$ represents the number of training samples within the target scene. The DSDAN we proposed is an end-to-end network designed to simultaneously process both source and target data streams. The architecture of DSDAN is depicted in Fig. 1. In essence, DSDAN incorporates a dual-stream feature extraction network, responsible for separately extracting features from the source and target scenes. Furthermore, it minimizes the disparities between different

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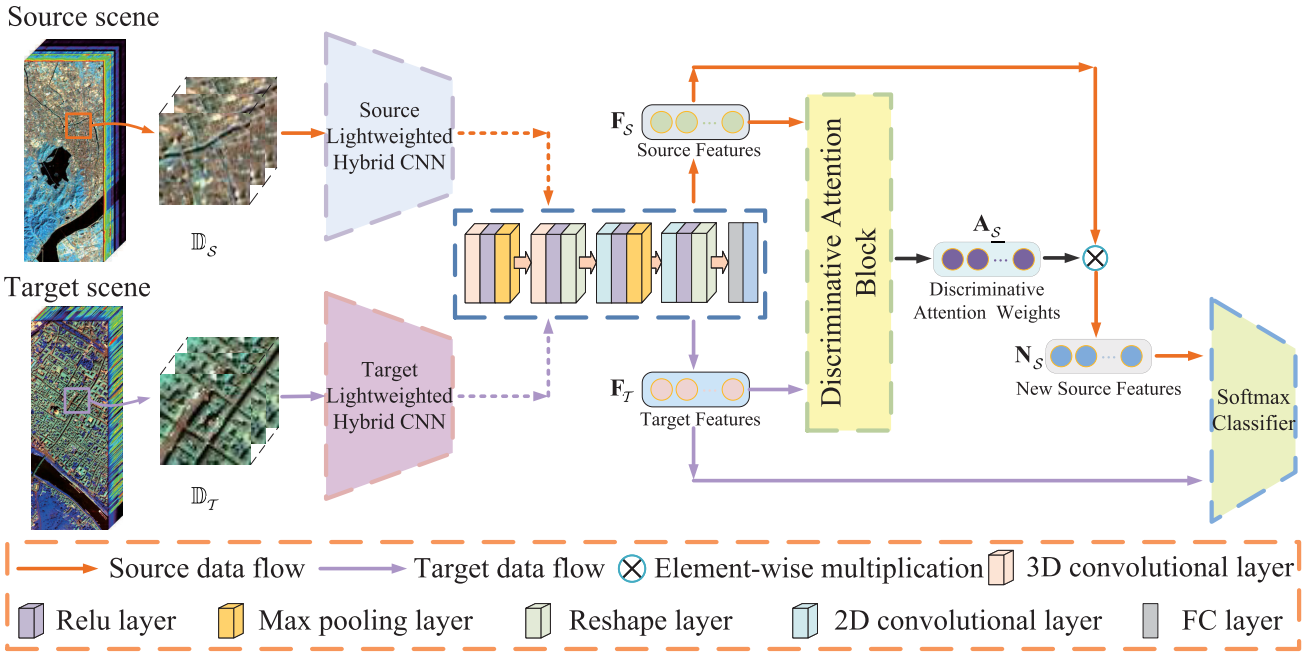


Fig. 1. Architecture of DSDAN.

scenes by introducing a DAB and imposing additional loss function constraints.

A. Dual-Stream Lightweight Hybrid CNN

CNN plays a crucial role in feature extraction due to its exceptional capability in effectively handling high-dimensional data. Nevertheless, when dealing with a limited number of training samples in HSIs, the adoption of overly complex models can result in underfitting issues. Fortunately, numerous algorithms have proven that the lightweight CNN architecture can produce outstanding results, even when the training samples are limited, as evidenced by studies such as [49], [50], [51], and so on.

Motivated by the models mentioned above, in this work, we introduce a DSLHC for feature extraction and alignment. As shown in Fig. 1, the DSLHC consists of both the source and target streams. Each stream allows the window of size $w \times w \times b$ as input. It is worth noting that similar land covers tend to exhibit spatial clustering. To facilitate convolutional operations, we typically expand the central pixel to a window of size $w \times w \times b$, with w representing the spatial dimension of each input window and b denoting the spectral dimension. The spectral dimension commonly differs between the source and target scenes. The two streams share an identical architecture, with specific parameter variations to accommodate their distinct input sample sizes. Specifically, the DSLHC employs 3-D convolutional layers as the first step to process input data, enabling the extraction of essential information across three dimensions, including height, width, and spectral dimension. Subsequently, the inclusion of a 3-D max pooling layer aids in reducing the spectral dimension within feature maps, effectively managing computational complexity while preserving vital information. Further refinement of advanced features is achieved through the iterative application of 3-D convolutional layers, followed by reshaping facilitated by a Reshape layer to seamlessly align with subsequent 2-D convolutional operations. The incorporation of 2-D convolutional layers further enhances spatial feature representation, with the assistance of ReLU activation and the 2-D max pooling layer to meticulously refine these features. Finally, two fully connected (FC) layers play a central role in mapping the features of the two distinct scenes to the same dimension for subsequent operations. The aligned source and target features extracted by DSLHC are represented as $\mathbf{F}_S \in \mathbb{V}^{n_S \times d}$ and $\mathbf{F}_T \in \mathbb{V}^{n_T \times d}$, where d signifies their feature dimension.

The functions of DSLHC can be summarized as follows.

1) Heterogeneous input sizes can be mapped to the same dimensional features through two designed separate networks.
2) The lightweight hybrid CNN is capable of efficiently extracting high-dimensional features with a limited number of parameters, effectively preventing overfitting.
3) DSLHC is an end-to-end network capable of feature modification through backpropagation, unlike pretrained networks.

B. Discriminative Attention Block

In cross-scene HSI classification, it is insufficient to solely project two scene features into the same dimension. It is also necessary to impose constraints on two scene features to further reduce domain differences. Specifically, following the successful alignment of features, our primary aim is to bolster the coherence between the features extracted from the source scene and the target scene. This means that between the two scenes, features belonging to the same class should be as similar as possible, while features from different classes should be as dissimilar as possible. Therefore, we introduce a DAB to address this issue, which enhances inter-class distinctiveness while preserving intraclass consistency through the incorporation of a cross-domain attention mechanism. Before

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providing a detailed description of DAB, we first introduce a formula for calculating the mean feature distance between the same classes in different scenes

$$U_{\text{same}}^{i,k} = \frac{1}{\sum_j \mu_{\mathcal{S},\mathcal{T},\text{same}}^{i,j}} \sum_j \mu_{\mathcal{S},\mathcal{T},\text{same}}^{i,j} \left| \mathbf{F}_{\mathcal{S}}^{i,k} - \mathbf{F}_{\mathcal{T}}^{j,k} \right| \tag{1}$$

where $\mathbf{F}_{\mathcal{S}}^{i,k}$ denotes the kth feature within the ith sample from the source scene, while $\mathbf{F}_{\mathcal{T}}^{j,k}$ represents the kth feature within the jth sample from the target scene and

$$\mu_{\mathcal{S},\mathcal{T},\text{same}}^{i,j} = \begin{cases} 1, & \text{class}(i) = \text{class}(j) \\ 0, & \text{class}(i) \neq \text{class}(j). \end{cases} \tag{2}$$

Similar to the previous formula, the mean feature distance between different classes in different scenes is calculated as follows:

$$U_{\text{diff}}^{i,k} = \frac{1}{\sum_j \mu_{\mathcal{S},\mathcal{T},\text{diff}}^{i,j}} \sum_j \mu_{\mathcal{S},\mathcal{T},\text{diff}}^{i,j} \left| \mathbf{F}_{\mathcal{S}}^{i,k} - \mathbf{F}_{\mathcal{T}}^{j,k} \right| \tag{3}$$

where

$$\mu_{\mathcal{S},\mathcal{T},\text{diff}}^{i,j} = \begin{cases} 1, & \text{class}(i) \neq \text{class}(j) \\ 0, & \text{class}(i) = \text{class}(j). \end{cases} \tag{4}$$

In the cross-domain attention mechanism, we aim to pay more attention to features that have smaller U_{same} but larger U_{diff}. Therefore, we calculate the kth discriminative attention weight within the ith sample from the source scene as follows:

$$a^{i,k} = \frac{U_{\text{diff}}^{i,k}}{U_{\text{same}}^{i,k}}. \tag{5}$$

The specific steps of DAB are listed in Algorithm 1. According to DAB, we obtain unique attention weights for each source feature. After performing element-wise multiplication between the source features $\mathbf{F}_{\mathcal{S}}$ and the discriminative attention weights $\mathbf{A}_{\mathcal{S}} \in \mathbb{V}^{n_{\mathcal{S}} \times d}$, we produce the new source features $\mathbf{N}_{\mathcal{S}} \in \mathbb{V}^{n_{\mathcal{S}} \times d}$, which assign different levels of attention to the source features to adapt to domain shift. The features that maintain interclass distinctiveness and intraclass consistency receive more attention, while others receive less attention. The new source features are subsequently employed to train the shared subnet alongside target features. The integration of these new source features significantly enhances the classification subnet training, leading to notable improvements in classification performance and, consequently, further alleviating the domain shift between the two scenes.

Algorithm 1 Discriminative Attention Block

Require:
Source features and target features $\mathbf{F}_{\mathcal{S}}, \mathbf{F}_{\mathcal{T}}$
Ensure:
New discriminative attention weights $\mathbf{A}_{\mathcal{S}}$
Initialize attention weights $\mathbf{A}_{\mathcal{S}} = [\alpha_1, \alpha_2, \ldots, \alpha_d] = 0$
for $i = 1$ **to** $n_{\mathcal{S}}$ **do**
  for $j = 1$ **to** $n_{\mathcal{T}}$ **do**
    for $k = 1$ **to** d **do**
      Calculate the mean feature distance between the same classes in different scenes $U_{same}^{i,k}$ with Eq. 1
      Calculate the mean feature distance between different classes in different scenes $U_{diff}^{i,k}$ with Eq. 3
      Update $\alpha_k = [a^{1,k}, \ldots, a^{i,k}]^{\mathbf{T}}$ with Eq. 5
    end for
  end for
end for
return $\mathbf{A}_{\mathcal{S}}$

C. Loss Function of DSDAN

In cross-scene HSI classification, it is of paramount importance to explore a variety of methods for enhancing domain connectivity and mitigating domain shift. The development of specific loss functions is indeed a viable approach. Zhu et al. [20] proposed a source cluster loss aimed at bringing samples of the same class closer together while pushing samples of different classes further apart. Deng et al. [52] leveraged the center magnet loss to project the source and target distributions into the same space, thereby inducing domain confusion. Ning et al. [53] formulated a contrastive loss that reduces the distance between the select sample and all other samples from different classes.

Motivated by the aforementioned loss functions, we design a CDL $\mathcal{L}_{\text{CDL}}$ to further reinforce domain invariance. Similar to the DAB, we first extend $U_{\text{same}}^{i,k}$ to the overall mean distance between same-class samples M_{same}

$$M_{\text{same}} = \frac{1}{\sum_{i,j} \mu_{\mathcal{S},\mathcal{T},\text{same}}^{i,j}} \sum_{i,j} \mu_{\mathcal{S},\mathcal{T},\text{same}}^{i,j} \left\| \mathbf{N}_{\mathcal{S}}^{i} - \mathbf{F}_{\mathcal{T}}^{j} \right\|_2. \tag{6}$$

Subsequently, the overall mean distance between different-class samples M_{diff} is defined as

$$M_{\text{diff}} = \frac{1}{\sum_{i,j} \mu_{\mathcal{S},\mathcal{T},\text{diff}}^{i,j}} \sum_{i,j} \mu_{\mathcal{S},\mathcal{T},\text{diff}}^{i,j} \left\| \mathbf{N}_{\mathcal{S}}^{i} - \mathbf{F}_{\mathcal{T}}^{j} \right\|_2. \tag{7}$$

It is worth noting that M_{same} and M_{diff} represent intraclass invariance and interclass difference. To transfer knowledge effectively, our goal is to minimize M_{same} and maximize M_{diff}. Therefore, the formulation of $\mathcal{L}_{\text{CDL}}$ is as follows:

$$\mathcal{L}_{\text{CDL}} = \frac{M_{\text{same}}}{M_{\text{diff}}}. \tag{8}$$

After formulating $\mathcal{L}_{\text{CDL}}$, we can obtain the complete DSDAN loss as follows:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}^{\mathcal{S}} + \mathcal{L}_{\text{CE}}^{\mathcal{T}} + \beta \mathcal{L}_{\text{CDL}} \tag{9}$$

where $\mathcal{L}_{\text{CE}}^{\mathcal{S}}$ and $\mathcal{L}_{\text{CE}}^{\mathcal{T}}$ represent the cross-entropy loss for the source and target classification tasks, respectively. β is a hyperparameter used to balance the classification task with domain invariance.

IV. Experiments

Three real cross-scene HSI datasets, namely the RPaviaU-DPaviaC dataset, the EHangzhou-RPaviaHR dataset, and the Houston dataset are employed to assess the effectiveness of the proposed DSDAN method. To meet real situations where source samples are sufficient while target samples are short, we randomly select 400 labeled source training samples per

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TABLE I
LAND-COVER CLASS AND COUNT OF LABELED SAMPLES IN THE RPAVIAU-DPAVIAC DATASET

Class		Count	
#	Name	RPaviaU	DPaviaC
1	Trees	3064	2424
2	Asphalt	6631	1704
3	Bitumen	1330	685
4	Shadow	947	241
5	Brick	3682	2237
6	Meadow	18649	1251
7	Soil	5029	1475

class and a mere five labeled target training samples per class. In the comparative analysis, we consider both single-scene HSI classification methods (hybrid spectral CNN (HSN) [54] and FTSCN [55]) as well as cross-scene HSI classification methods (deep cross-domain FSL (DCFSL) [56], domain-invariant attention network (DIAN) [57], single-source domain expansion network (SDEnet) [58], GMTN [19]). Subsequently, performance evaluation metrics such as class-specific accuracy (CSA), overall accuracy (OA), average accuracy (AA), kappa coefficient (k), Params, and execution time are utilized to gauge the performance of these algorithms. Furthermore, to reduce the randomness in experiments and enhance their credibility, we conduct repeated experiments using 5 sets of random samples. Ultimately, we report the average and standard deviation values of the results to provide a robust and reliable assessment of method performance.

A. Experimental Datasets

1) *RPaviaU-DPaviaC Dataset:* The RPaviaU-DPaviaC dataset is constructed by amalgamating two publicly accessible HSI datasets: the ROSIS Pavia University (RPaviaU) scene and the DAIS Pavia Center (DPaviaC) scene. The RPaviaU dataset, featuring dimensions of 610 × 340 × 103, was acquired by the ROSIS HSI sensor over the terrain of the University of Pavia, Italy. Conversely, the DPaviaC dataset, with dimensions of 400 × 400 × 72, was collected using the DAIS sensor over the central area of Pavia City, Italy. These two scenes share a common set of seven land cover classes. Detailed information about the number of labeled samples in each land cover class is presented in Table I. Moreover, visual representations of the false color images and the corresponding ground-truth maps are thoughtfully displayed in Fig. 2.

2) *EHangzhou-RPaviaHR Dataset:* The EHangzhou-RPaviaHR dataset encompasses two distinctive scenes: the EO-1 Hangzhou (EHangzhou) scene and the ROSIS Pavia HR (RPaviaHR) scene. The EHangzhou scene, sourced from the EO-1 Hyperion hyperspectral sensor, was obtained over Hangzhou City, Zhejiang, China, and possesses dimensions of 590 × 230 × 198 as reported by Ye et al. [9]. In contrast, the target scene, RPaviaHR, was captured using the ROSIS HSI sensor over Pavia city, Italy, and is characterized by dimensions of 1400 × 512 × 102. These two scenes share a common subset of three land cover classes, with comprehensive details provided in Table II. Evidently, the false color images and the associated ground-truth maps are thoughtfully presented for visual reference in Fig. 3.

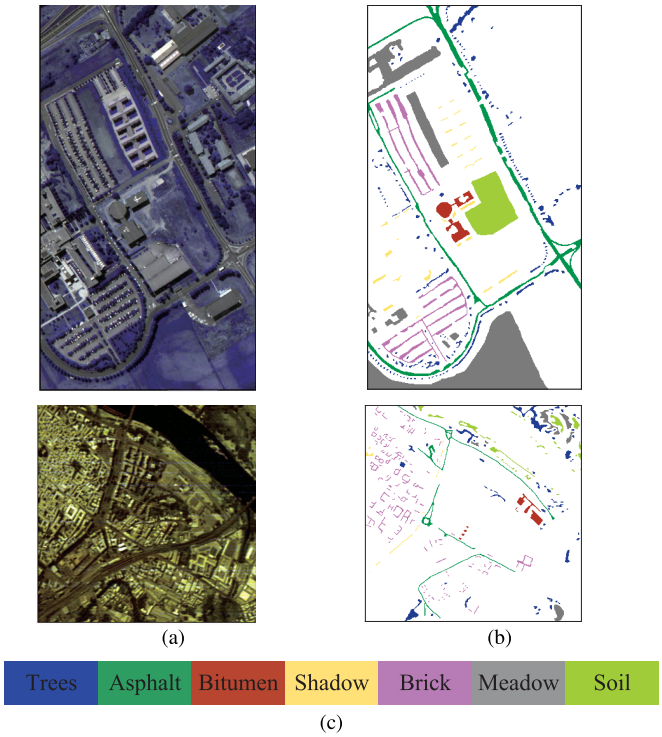


Fig. 2. Images within the RPaviaU-DPaviaC dataset. The upper image represents RPaviaU, while the lower represents DPaviaC. (a) False color images. (b) Ground-truth maps. (c) Seven shared land-cover classes.

TABLE II
LAND-COVER CLASS AND COUNT OF LABELED SAMPLES IN THE EHANGZHOU-RPAVIAHR DATASET

Class		Count	
#	Name	EHangzhou	RPaviaHR
1	Water	18403	22525
2	Ground/Building	77450	145416
3	Vegetation	40207	22961

3) *Houston Dataset:* The Houston dataset encompasses two distinct scenes: the Houston2013 and the Houston2018. The former, originating from the airborne visible/infrared imaging spectrometer (AVIRIS), captures the urban landscape of Houston, Texas, USA, with dimensions of 349 × 1905 × 144, and serves as the source scene. In contrast, the target scene Houston2018, building on its predecessor, incorporates LiDAR data for the identical geographical locale. The spatial extent of Houston2018 is delineated as 601 × 2384 × 48. These scenes share seven common land cover classes, with exhaustive details provided in Table III. Additionally, for visual elucidation, false color images alongside their corresponding ground-truth maps are elaborately presented in Fig. 4.

These three datasets thoroughly consider practical factors and are valuable resources for assessing the cross-scene HSI classification problem.

B. Experiments on Parameter Tuning

This section delves into the influence of parameter variations on the performance of DSDAN using three datasets.

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TABLE III
LAND-COVER CLASS AND COUNT OF LABELED SAMPLES IN THE HOUSTON DATASET

	Class	Count	
#	Name	Houston 2013	Houston 2018
1	Grass healthy	1251	9799
2	Grass stressed	1254	32502
3	Trees	1244	18636
4	Water	325	266
5	Residential buildings	1268	39762
6	Non-residential buildings	1244	223684
7	Road	1250	45810

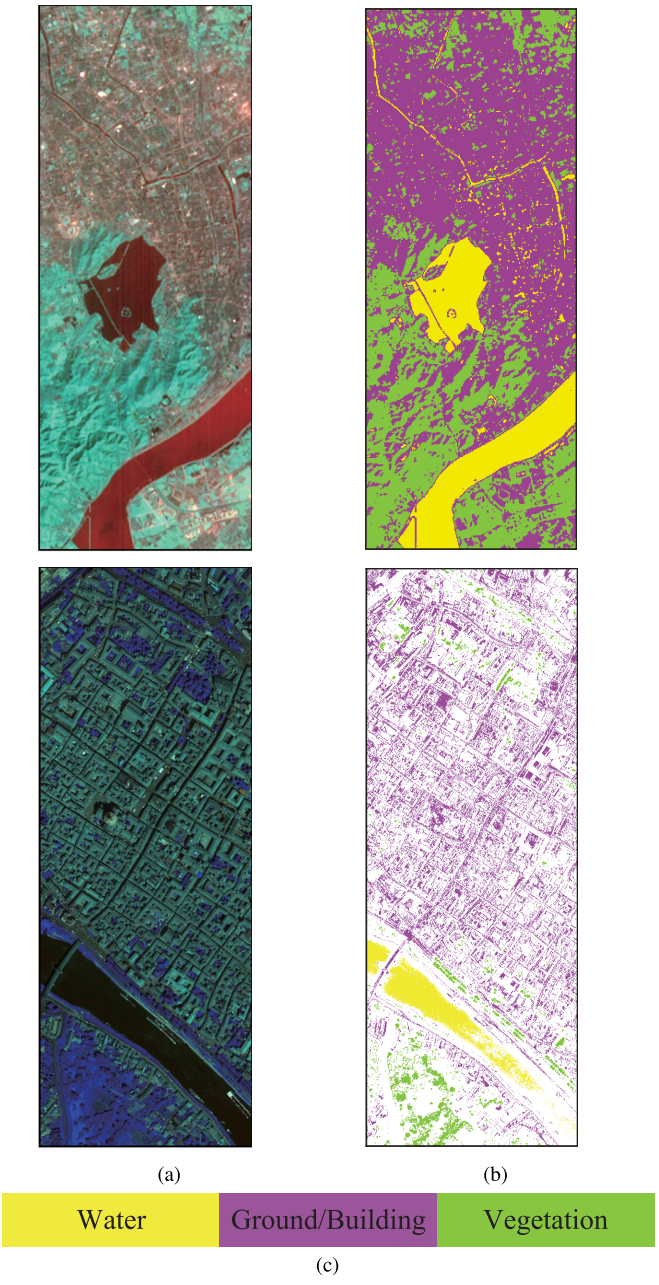


Fig. 3. Images within the EHangzhou-RPaviaHR dataset. The upper image represents EHangzhou, while the lower represents RPaviaHR. (a) False color images. (b) Ground-truth maps. (c) Three shared land cover classes.

The parameters under consideration include the feature alignment dimension d, the balancing hyperparameter β, the base learning rate lr, and the patch size $w \times w$. The discussion regarding these hyperparameters is as follows.

1) *Feature Alignment Dimension d:* The choice of d is in $\{128, 256, \ldots, 1024\}$. As demonstrated in Table IV, it becomes apparent that opting for a feature alignment dimension that is either too small or too large can pose challenges for effective transfer learning. The optimal result is attained when d is set to 256.
2) *Balancing Hyperparameter β:* As shown in Table IV, the hyperparameter β is selected from the range $\{0.2, 0.4, \ldots, 1.0\}$. A small value of β can result in the classification task dominating, thereby burdening the loss function. Conversely, an excessively large value of β can cause DSDAN to overly emphasize the loss function, hindering effective feature extraction. The most favorable outcome is observed when $\beta = 0.6$.
3) *Base Learning Rate lr:* The value of lr is chosen from the sets $\{10^{-5}, 10^{-4}, \ldots, 10^{-1}\}$. As detailed in Table IV, opting for a learning rate that is excessively small can result in the training process fixating on local optima, whereas an overly large learning rate may introduce instability in the training process and preclude the attainment of optimal results. Consequently, the most suitable configuration for the learning rate in this study is determined to be 10^{-3}, ensuring a balanced and effective training process.
4) *Patch Size $w \times w$:* The value of $w \times w$ is selected from the sets $\{3 \times 3, 5 \times 5, \ldots, 11 \times 11\}$. As elaborated in Table IV, insufficient patch size may result in inadequate information for each input, whereas excessive patch size may entail the inclusion of extensive information from other classes in each patch, thereby exacerbating the challenge of domain adaptation due to the disparate distribution of land cover classes between two scenes. The most suitable configuration for the patch size in this study is determined to be 5×5.

To summarize, we adopt the parameter setting of ($d = 256, \beta = 0.6, \text{lr} = 10^{-3}, w \times w = 5 \times 5$) for subsequent experiments.

C. Ablation Study

To assess the contribution of each proposed component, we conduct ablation experiments in this section. The specific descriptions of each method are as follows.

1) *Target-Stream Lightweight Hybrid CNN (TSLHC):* TSLHC is a single-scene classification algorithm proposed in this work. It simply incorporates fundamental 3-D, 2-D, and max-pooling operations. The lightweight architecture is designed to reduce network complexity while preserving efficacy.
2) *DSLHC:* DSLHC extends TSLHC by integrating two streams for the classification task. Initially, alignment features are extracted from two scenes, and these features are subsequently employed to train a shared subnet for classification.
3) *Dual-Stream Attention Network (DSAN):* DSAN is the combination of DSLHC with DAB. With the assistance

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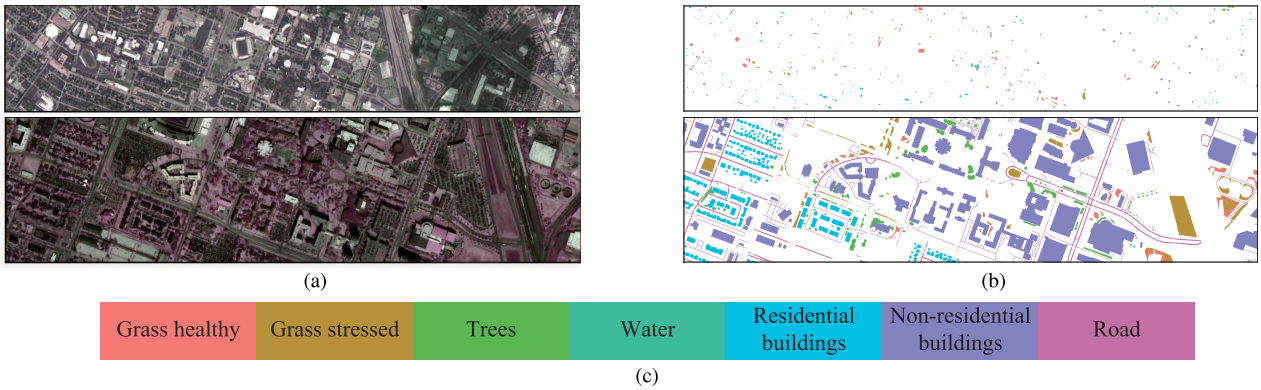


Fig. 4. Images within the Houston dataset. The upper image represents Houston 2013, while the lower represents Houston 2018. (a) False color images. (b) Ground-truth maps. (c) Seven shared land cover classes.

TABLE IV
CLASSIFICATION ACCURACIES BETWEEN DIFFERENT PARAMETERS

	RPaviaU-DPaviaC			EHangzhou-RPaviaHR			Houston		
d	OA	AA	κ	OA	AA	κ	OA	AA	κ
128	0.8986	0.8970	0.8769	0.9261	0.9676	0.8315	0.8433	0.8938	0.7568
256	**0.9358**	**0.9351**	**0.9219**	**0.9996**	**0.9995**	**0.9989**	**0.8614**	**0.8981**	**0.7807**
512	0.8873	0.8563	0.8629	0.9828	0.9923	0.9573	0.8399	0.8934	0.7529
768	0.8743	0.8878	0.8482	0.9621	0.9832	0.9089	0.8508	0.8872	0.7653
1024	0.8698	0.8734	0.8423	0.9273	0.9681	0.8340	0.8316	0.8838	0.7394
β	OA	AA	κ	OA	AA	κ	OA	AA	κ
0.2	0.8617	0.8481	0.8316	0.9258	0.9647	0.8327	0.7957	0.8394	0.6821
0.4	0.8712	0.8749	0.8439	0.9390	0.9714	0.8587	0.8275	0.8803	0.7322
0.6	**0.9358**	**0.9351**	**0.9219**	**0.9996**	**0.9995**	**0.9989**	**0.8614**	**0.8981**	**0.7807**
0.8	0.8912	0.8725	0.8682	0.9513	0.9609	0.8857	0.8446	0.8840	0.7550
1.0	0.8668	0.8830	0.8390	0.9197	0.9647	0.8187	0.8272	0.8881	0.7348
lr	OA	AA	κ	OA	AA	κ	OA	AA	κ
10^{-5}	0.8280	0.8352	0.7920	0.8436	0.9315	0.6971	0.7848	0.8714	0.6779
10^{-4}	0.8604	0.8779	0.8322	0.9622	0.9833	0.9133	0.8264	0.8864	0.7248
10^{-3}	**0.9358**	**0.9351**	**0.9219**	**0.9996**	**0.9995**	**0.9989**	**0.8614**	**0.8981**	**0.7807**
10^{-2}	0.8686	0.8685	0.8412	0.9148	0.9626	0.8089	0.8196	0.8918	0.7257
10^{-1}	0.8174	0.7954	0.7797	0.8828	0.9485	0.7564	0.8086	0.8770	0.7075
$w \times w$	OA	AA	κ	OA	AA	κ	OA	AA	κ
3×3	0.8636	0.8630	0.8343	0.9725	0.9878	0.9331	0.8034	0.8716	0.6997
5×5	**0.9358**	**0.9351**	**0.9219**	**0.9996**	**0.9995**	**0.9989**	**0.8614**	**0.8981**	**0.7807**
7×7	0.8978	0.8830	0.8759	0.9551	0.9802	0.8957	0.8437	0.8960	0.7573
9×9	0.8675	0.8715	0.8400	0.9739	0.9867	0.9362	0.8581	0.8963	0.7765
11×11	0.8260	0.8474	0.7918	0.9592	0.9786	0.9019	0.7606	0.8678	0.7592

of DAB, source features, which receive different attention, can be brought closer to the target features.

4) *DSDAN:* DSDAN is the whole network we proposed, which contains the DSAN and CDL. By incorporating CDL after DSAN, the features from the two scenes undergo a more pronounced alignment, resulting in a more effective reduction of domain shift.

The results of the three datasets are presented in Table V. It is proven that with the assistance of source information, DSLHC exhibits higher accuracy compared to TSLHC, and both DAB and CDL undeniably mitigate domain shifts, thereby facilitating the process of transfer learning.

D. Comparisons and Analyses

For the RPaviaU-DPaviaC dataset, the classification performance of each method is summarized in Table VI. Classification maps can be seen in Fig. 5.

The summary of RPaviaU-DPaviaC dataset results is as follows.

1) Without knowledge from the source scene, single-scene algorithms (HSN and FTSCN) exhibit poor performance compared to cross-scene algorithms, emphasizing the pivotal role of cross-scene algorithms. Among them, HSN performs inferior performance, likely due to overfitting, which results from a shortage of target training samples, preventing effective training of the large-scale network. FTSCN stands out among single-scene algorithms, highlighting the effectiveness of attention mechanisms in mitigating the small-sample-size problem.
2) Among all algorithms, DSDAN consistently achieves the highest classification performance, simultaneously showcasing the most consistent results across individual classes. Unlike other cross-scene algorithms (DCFSL, DIAN, SDEnet, and GMTN), which may experience abrupt drops in accuracy for specific classes, DSDAN consistently upholds CSA levels at approximately, thereby demonstrating the efficacy of the cross-domain

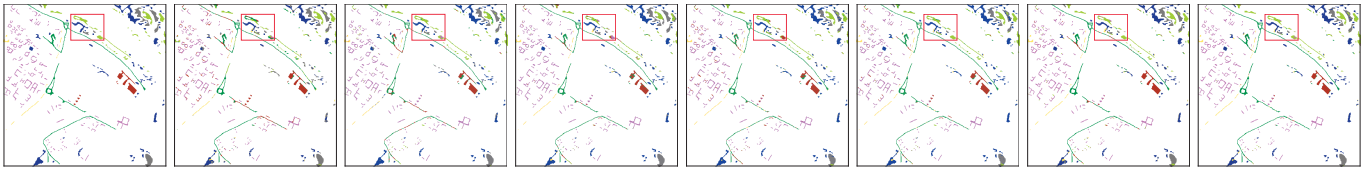
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TABLE V
ABLATION STUDY ON THE THREE DATASETS

Method	Cross scene	DAB	CDL	RPaviaU-DPaviaC OA	RPaviaU-DPaviaC AA	RPaviaU-DPaviaC κ	EHangzhou-RPaviaHR OA	EHangzhou-RPaviaHR AA	EHangzhou-RPaviaHR κ	Houston OA	Houston AA	Houston κ
TSLHC				0.8314	0.8558	0.7983	0.8721	0.9437	0.7389	0.7792	0.8245	0.6488
DSLHC	✓			0.8876	0.8887	0.8641	0.8935	0.9532	0.7739	0.8045	0.8213	0.6821
DSAN	✓	✓		0.9060	0.9164	0.8864	0.9508	0.9782	0.8879	0.8218	0.8926	0.7271
DSDAN	✓	✓	✓	**0.9358**	**0.9351**	**0.9219**	**0.9996**	**0.9995**	**0.9989**	**0.8614**	**0.8981**	**0.7807**

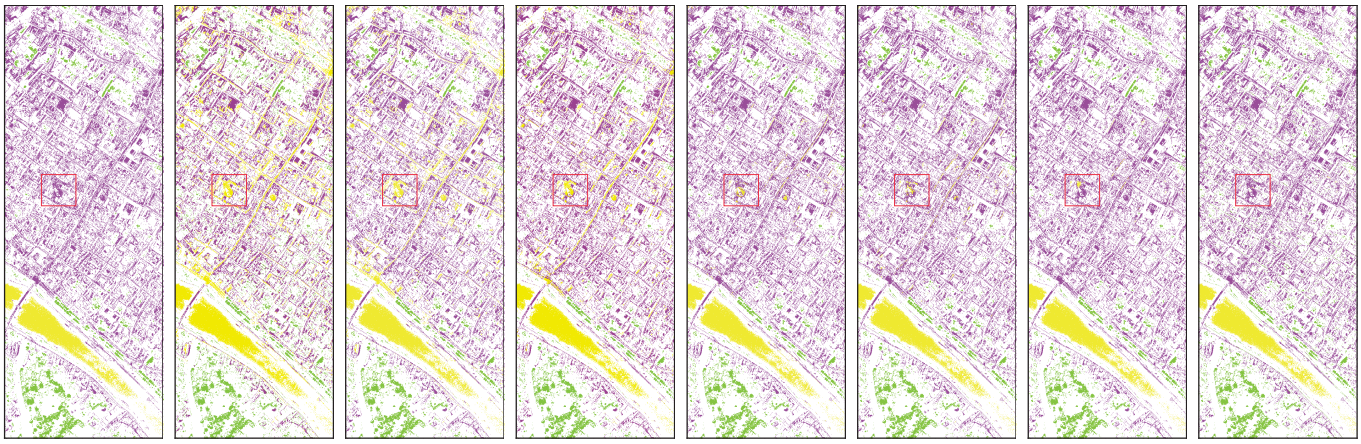
TABLE VI
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS FOR THE RPAVIAU-DPAVIAC DATASET

Class	HSN [54]	FTSCN [55]	DCFSL [56]	DIAN [57]	SDEnet [58]	GMTN [19]	DSDAN
1	0.6656	0.8284	0.8818	0.8991	0.9243	0.9235	**0.9252**
2	0.7528	0.8046	**0.9570**	0.8387	0.8458	0.8987	0.8953
3	0.9162	0.8779	0.8824	**0.9191**	0.8985	0.8389	0.8934
4	0.6568	0.9364	**0.9958**	0.9873	0.9746	0.8635	0.9789
5	0.8347	0.8902	0.9431	0.9453	0.9418	0.9486	**0.9564**
6	0.9358	0.7689	0.9920	0.8491	**0.9759**	0.9176	0.9374
7	0.6735	0.9259	0.7177	**0.9524**	0.9020	0.9404	0.9180
OA	0.7700 ± 0.0281	0.8510 ± 0.0271	0.9006 ± 0.0101	0.9042 ± 0.0072	0.9175 ± 0.0097	0.9129 ± 0.0187	**0.9267 ± 0.0129**
AA	0.7765 ± 0.0304	0.8618 ± 0.0182	0.9100 ± 0.0099	0.9130 ± 0.0069	0.9233 ± 0.0101	0.8994 ± 0.0092	**0.9292 ± 0.0083**
k	0.7253 ± 0.0375	0.8204 ± 0.0239	0.8794 ± 0.0130	0.8841 ± 0.0085	0.9000 ± 0.0112	0.9065 ± 0.0148	**0.9110 ± 0.0155**



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 5. RPaviaU-DPaviaC dataset. (a) Ground truth. (b) HSN (0.7814). (c) FTSCN (0.8622). (d) DCFSL (0.9079). (e) DIAN (0.9093). (f) SDEnet (0.9189). (g) GMTN (0.9237). (h) DSDAN (0.9358).



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 6. EHangzhou-RPaviaHR dataset. (a) Ground truth. (b) HSN (0.8340). (c) FTSCN (0.8992). (d) DCFSL (0.9397). (e) DIAN (0.9608). (f) SDEnet (0.9791). (g) GMTN (0.9727). (h) DSDAN (0.9996).

attention mechanism in effectively mitigating domain adaptation challenges.

3) Table VII indicates that, due to the incorporation of source scene data, cross-scene algorithms generally exhibit longer runtime and involve more parameters compared to single-scene algorithms. The DSDAN achieves superior performance while maintaining lightweight characteristics.

Following an in-depth analysis of the RPaviaU-DPaviaC dataset, we transition to conducting experiments on the second dataset, EHangzhou-RPaviaHR. The resulting outcomes are meticulously presented in Table VIII, while the corresponding classification maps are visually portrayed in Fig. 6. Herein, we provide a comprehensive account of the specific findings and in-depth analyses.

1) Similar to the results of the RPaviaU-DPaviaC dataset, the performance of single-scene algorithms is notably inferior to that of cross-scene algorithms due to the absence of source scene knowledge.

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TABLE VII

PARAMS, RUNNING TIME, AND TESTING TIME OF EACH METHOD ON DIFFERENT DATASETS

Datasets	Methods	HSN [54]	FTSCN [55]	DCFSL [56]	DIAN [57]	SDEnet [58]	GMTN [19]	DSDAN
RPaviaU-DPaviaC	Params (M)	7.28	1.64	4.24	3.28	2.64	4.52	3.54
	Running Time (s)	1062	762	1297	1093	956	1892	1368
	Testing Time (s)	12	9	18	14	16	27	10
EHangzhou-RPaviaHR	Params (M)	7.28	1.69	4.26	3.88	2.75	4.74	3.79
	Running Time (s)	1129	851	1496	1467	1169	1959	1461
	Testing Time (s)	67	51	149	203	238	419	133
Houston	Params (M)	7.28	1.73	4.29	4.11	2.92	5.06	3.91
	Running Time (s)	1397	939	1533	1576	1192	2099	1529
	Testing Time (s)	179	136	273	529	570	762	197

TABLE VIII

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS FOR THE EHANGZHOU-RPAVIAHR DATASET

Class	HSN [54]	FTSCN [55]	DCFSL [56]	DIAN [57]	SDEnet [58]	GMTN [19]	DSDAN
1	0.9848	**0.9998**	0.9993	0.9988	0.9989	0.9958	0.9995
2	0.7894	0.8586	0.9144	0.9408	0.9703	0.9651	**0.9827**
3	0.9981	0.9970	0.9944	0.9887	0.9548	0.9978	**0.9991**
OA	0.8293 ± 0.0756	0.8919 ± 0.0094	0.9341 ± 0.0273	0.9534 ± 0.0011	0.9718 ± 0.0162	0.9727 ± 0.0186	**0.9867 ± 0.0273**
AA	0.9248 ± 0.0325	0.9518 ± 0.0036	0.9694 ± 0.0095	0.9761 ± 0.0149	0.9747 ± 0.0190	0.9862 ± 0.0084	**0.9938 ± 0.0196**
k	0.6688 ± 0.0152	0.7653 ± 0.0170	0.8489 ± 0.0056	0.8888 ± 0.0057	0.9306 ± 0.0373	0.9363 ± 0.0181	**0.9670 ± 0.0305**

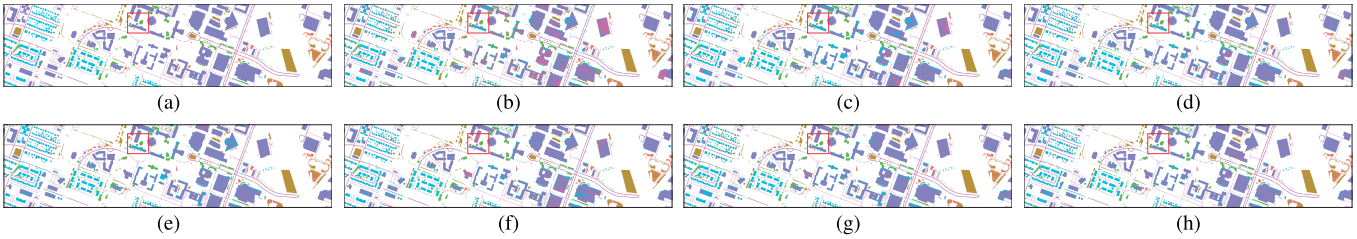


Fig. 7. Houston dataset. (a) Ground truth. (b) HSN (0.7599). (c) FTSCN (0.8135). (d) DCFSL (0.8360). (e) DIAN (0.8397). (f) SDEnet (0.8459). (g) GMTN (0.8497). (h) DSDAN (0.8614).

TABLE IX

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS FOR THE HOUSTON DATASET

Class	HSN [54]	FTSCN [55]	DCFSL [56]	DIAN [57]	SDEnet [58]	GMTN [19]	DSDAN
1	0.9196	0.9587	0.9422	0.9311	0.9588	0.9462	**0.9628**
2	0.8987	0.9189	**0.9355**	0.9299	0.9147	0.9023	0.9045
3	0.8370	0.8881	0.8976	0.8806	0.9031	**0.9306**	0.9052
4	0.9940	0.9880	0.9940	0.9934	0.9880	0.9819	**0.9994**
5	0.8077	0.7912	0.8317	0.7652	0.7929	0.7998	**0.8431**
6	0.6982	0.7790	0.8089	0.8278	0.8324	0.8235	**0.8392**
7	0.7760	0.7771	0.8180	0.8063	0.8239	**0.8296**	0.8241
OA	0.7487 ± 0.0329	0.8026 ± 0.0209	0.8316 ± 0.0088	0.8328 ± 0.0087	0.8412 ± 0.0037	0.8373 ± 0.0165	**0.8501 ± 0.0074**
AA	0.8473 ± 0.0297	0.8715 ± 0.0132	0.8897 ± 0.0107	0.8763 ± 0.0094	0.8878 ± 0.0028	0.8877 ± 0.0077	**0.8970 ± 0.0097**
k	0.6323 ± 0.0413	0.6979 ± 0.0254	0.7389 ± 0.0112	0.7381 ± 0.0167	0.7519 ± 0.0049	0.7462 ± 0.0191	**0.7649 ± 0.0112**

2) Classifying Ground/Building (class 2) with other classes is a challenging task. However, DSDAN outperforms other cross-scene algorithms, demonstrating its effectiveness for cross-scene classification.

Similar to the preceding two datasets, the classification results for the final dataset, Houston, are presented in Table IX, accompanied by corresponding classification maps displayed in Fig. 7. The following provides a detailed analysis of this dataset.

1) Unlike the previous two datasets, Houston 2018 exhibits a challenge in that numerous classes are densely concentrated in a small area, making their separation more challenging. DSDAN excels in capturing the distinctions between them through its cross-domain attention mechanism, resulting in optimal classification performance.
2) Because DAB in DSDAN operates on source scene features, this results in faster testing speeds during the classification process.

After a comprehensive evaluation of all conducted experiments, the DSDAN we proposed achieves the best performance compared with other algorithms. However, the DSDAN still has certain limitations as follows.

1) The DSDAN necessitates a small number of training samples from the target scene, which does not suit the situation when the target scene has no labeled samples in some classes.
2) Initially designed to address the challenges associated with limited sample sizes, the DSDAN may not achieve optimal performance in situations characterized by an

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abundance of training samples when compared with other algorithms.

In subsequent works, we will strive to address these issues to enhance the model's generalizability.

V. Conclusion

In this work, we propose a novel approach for heterogeneous transfer learning named DSDAN. Our primary objective is to tackle the challenges associated with small-sample-size issues in cross-scene HSI classification. DSDAN consists of three key components. First, DSLHC is developed to extract aligned spatial–spectral features from two distinct input data sizes. Furthermore, DAB maintains domain invariance by selectively attending to the source features extracted from DSLHC. Finally, CDL is employed to further mitigate domain shift. The efficacy of DSDAN is empirically demonstrated through experiments conducted on three authentic cross-scene HSI datasets.

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